UZHAVAN AI: A TAMIL-FIRST MULTIMODAL AGRICULTURAL INTELLIGENCE AND DECISION-SUPPORT PLATFORM FOR TAMIL NADU FARMERS

SDG : 2 – Zero Hunger

PROJECT REQUIREMENT SPECIFICATION REPORT (23AI2014)

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE AND ENGINEERING
(AI/AIML)

Academic Year 2026-2027
ODD SEMESTER

DIVISION OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
SCHOOL OF COMPUTER SCIENCE AND TECHNOLOGY

Karunya INSTITUTE OF TECHNOLOGY AND SCIENCES
(Declared as Deemed to be University under Sec.3 of the UGC Act, 1956)
MoE, UGC & AICTE Approved
NAAC A++ Accredited

November 2026

BONAFIDE CERTIFICATE

This is to certify that the project requirement specification report entitled **"UZHAVAN AI: A Tamil-First Multimodal Agricultural Intelligence and Decision-Support Platform for Tamil Nadu Farmers"** is a bonafide work done during the odd semester of the academic year 2026-2027 by **AJAI KUMAR N (URK23CS5064)**

in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (AI /AIML) of Karunya Institute of Technology and Sciences.

Signature of the HOD

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Submitted for the Viva Voce held on 10/11/2026

DECLARATION

We, hereby affirm our commitment to the development and implementation of the project described in this report. We are responsible for the creation and development of the project outlined in this Project Requirement Specification (PRS) report, hereby declare the following:

1. We have thoroughly reviewed and analyzed the requirements and have ensured that they are accurately represented in this report to the best of our knowledge.
2. We commit to adhering to the specifications, functionalities, and features outlined in this PRS during the development phase of the project.
3. We will strive to implement industry best practices, coding standards, and security measures to develop a high-quality, reliable, and secure software solution.
4. We affirm our commitment to delivering the solution within the agreed-upon timelines, while prioritizing quality.
5. We jointly undertake to work towards achieving the selected project outcome (Patent / Startup / Consultancy / Journal Publication) committed to under this project.



Signature of student



Signature of the Supervisor

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1. INTRODUCTION

1.1 Background

Agriculture remains the main source of livelihood for a large share of households in Tamil Nadu. A farmer makes decisions continuously across the season: which crop to sow, when to prepare the land, when to irrigate, how much fertiliser to apply, how to respond to a pest or disease, and when and where to sell the produce. Each of these decisions depends on the crop, its growth stage, the district, the season and the current weather, and each draws on information that is held in a different place.

Useful information already exists. Tamil Nadu Agricultural University (TNAU) publishes crop-production guides, pest and disease notes and soil recommendations. Weather services publish forecasts, AGMARKNET publishes market arrivals and prices, soil-information products and Soil Health Cards describe nutrient status, and government portals list agricultural schemes. Research datasets of crop-leaf images and multi-sensor satellite imagery have made disease screening and crop monitoring technically possible. The difficulty for a farmer is not the absence of information but the effort needed to locate, interpret and combine it for one specific field and one specific question.

1.2 Motivation

Most digital agricultural tools address a single need. A weather application gives a forecast without knowing the crop stage; a disease application recognises symptoms but offers little management guidance; a market portal shows prices in technical tables; and general-purpose chatbots answer fluently but may invent dosages, prices or weather values. Many of these tools are also English-first and text-heavy, which limits their usefulness for farmers who prefer Tamil, Tanglish or speech.

Large language models (LLMs) make natural-language interaction possible, but on their own they are not a safe source of agricultural facts. The motivation for this project is therefore to build a system in which the language model only **explains** evidence that has first been retrieved from trusted sources, obtained from current data providers or produced by a specialised model, and in which uncertainty and missing evidence are shown honestly to the farmer.

1.3 About the Project

UZHAVAN AI (“Uzhavan” means farmer in Tamil) is a Tamil-first agricultural intelligence and decision-support platform for farmers in Tamil Nadu. It consists of a Flutter Android application and a FastAPI backend that coordinates authentication, farmer and farm context, a retrieval-augmented agricultural assistant, and information modules for weather, market prices, soil and government schemes. A research-stage crop-leaf disease-screening model (EfficientNetV2-S) and a satellite crop-intelligence workstream based on the SICKLE dataset are integrated behind explicit release gates.

The central design idea is an **evidence-first, capability-based workflow**: a farmer’s request is interpreted, only the capabilities it needs are invoked, evidence is gathered and validated, and only then is a locally hosted language model (Qwen2.5 7B Instruct served through Ollama) asked to explain that evidence in farmer-friendly language. No paid cloud AI service or cloud speech service is used; language generation, vector retrieval and speech run locally.

1.4 Sustainable Development Goal Alignment

The project is aligned primarily with **SDG 2 – Zero Hunger**, in particular Target 2.3 (improve the productivity and incomes of small-scale food producers) and Target 2.4 (promote sustainable and resilient agricultural practices). It also supports SDG 6 (Clean Water and Sanitation) through irrigation-relevant weather information, SDG 13 (Climate Action) through weather-aware advice, and SDG 15 (Life on Land) through soil-health information. These are intended contribution areas; measured improvements in yield, income or resource use have not yet been demonstrated and require a field study.

1.5 Organisation of the Report

Section 2 defines the problem, scope and objectives. Section 3 surveys related work and identifies the research gap. Section 4 analyses the functional, non-functional, data and interface requirements. Section 5 presents the proposed methodology and system design. Section 6 describes the implementation, and Section 7 reports the results obtained so far with a discussion of their limits. Section 8 concludes the report and outlines future work, followed by the references and Annexure I, which records the evidence for the selected project outcome (journal publication).

2. PROBLEM DEFINITION, SCOPE AND OBJECTIVES

2.1 Problem Context

The information that a Tamil Nadu farmer needs is distributed across agricultural university publications, government portals and schemes, weather providers, market-price systems, soil surveys and soil-test records, crop-disease image datasets, satellite research datasets and the farmer's own knowledge of the crop, location, land area and crop stage. This fragmentation produces four practical problems.

2.1.1 Information fragmentation

A single contextual question, such as whether fertiliser can be applied to paddy before tomorrow's expected rain, may require several applications, websites or offices. A generic answer does not take the farmer's crop, district, crop stage or present weather into account.

2.1.2 Language accessibility

Agricultural material often uses technical English or complex interfaces that are inconvenient for farmers who prefer Tamil, Tanglish or voice interaction.

2.1.3 Context gap

Agronomic guidance depends on location, crop, crop stage, farm area, season and current conditions. A recommendation for one crop or stage must not be applied to another – for example, a paddy fertiliser rate must not be given to a farmer whose active farm is banana.

2.1.4 Trust and safety

Language models and image classifiers can produce unsupported or incorrect output. A safe agricultural assistant must distinguish retrieved evidence, current provider observations, calculated values, model predictions, general guidance and unavailable information, and it must refuse or qualify high-risk instructions – especially exact chemical dosages – when applicable evidence is missing.

2.2 Problem Statement

How can heterogeneous agricultural information, AI models and current farm context be integrated into a Tamil-first mobile platform that provides accessible, traceable and safety-aware decision support to farmers in Tamil Nadu?

2.3 Scope

2.3.1 In scope

- Farmer login and session management through the existing OTP/JWT authentication flow.
- Farmer profile, farm information and farm-diary records.
- A Flutter Android interface with Home, Ask AI, Weather, Market, Soil, Schemes, Scan, Farm and Profile screens.
- A text-based agricultural assistant with TNAU-oriented retrieval-augmented generation (RAG) and source attribution, using local Ollama-based generation.

- Weather and forecast display; market observations with units and snapshot/freshness labels; regional soil information and user-provided soil measurements; government-scheme retrieval.
- Research-stage disease-image screening behind a safety gate, and a satellite research and validation-gate framework.
- Evidence provenance, uncertainty display, safe fallback behaviour and local device text-to-speech.

2.3.2 Out of scope

- Production-certified crop disease diagnosis, or unrestricted disease prediction for all crops and field conditions.
- Farmer-facing satellite prediction before licence, unit, geographic-scope and independent-evaluation gates have passed.
- Exact pesticide or chemical dosage without applicable, trusted source evidence.
- Cloud speech-to-text/text-to-speech or paid hosted AI inference as a fallback.
- Guaranteed statewide yield prediction, and claims of increased income or yield without field evidence.
- Large-scale deployment and farmer-centred field evaluation, which are planned as later work.

2.4 Objectives

1. Develop a Flutter mobile application for farmer-facing agricultural information access.
2. Provide Tamil-oriented and English interaction, with Tanglish support as a design objective.
3. Maintain farmer and farm context (district, crop, crop stage, land area) where supplied by the user.
4. Integrate local language generation through Ollama without paid hosted LLM APIs.
5. Retrieve agricultural knowledge from a TNAU-oriented corpus stored in ChromaDB.
6. Distinguish source-grounded answers from general guidance and uncertain responses.
7. Integrate weather, market, soil and government-scheme information with source and freshness labels.
8. Implement research-stage disease-image screening with confidence and uncertainty (UNSURE) controls.
9. Explore satellite-derived crop and crop-calendar intelligence subject to data and model release gates.
10. Apply grounding and safety checks, especially for disease-related advice and chemical dosage.
11. Provide spoken responses using local device TTS where the required language voice is installed.
12. Evaluate the system through software tests, API tests and physical-device tests, followed in later work by expert and farmer studies.

2.5 Intended Users

Table 2.1: User classes of UZHAVAN AI

User class	Characteristics and needs
Farmers in Tamil Nadu	Primary users; need quick, clear and practical guidance, may prefer Tamil or speech, and may have limited technical literacy.
Agricultural officers and experts	Escalation and domain-review stakeholders; the system supports but does not replace their judgement.
Researchers and students	Examine the architecture, datasets, model limitations and evaluation.
Administrators / operators	Maintain data sources, infrastructure, configuration and validation records.

3. LITERATURE SURVEY

This section reviews work in the individual areas that UZHAVAN AI brings together: satellite crop intelligence, crop-disease recognition, agricultural knowledge retrieval and advisory systems, weather/soil/market information, and speech for Indian languages.

3.1 Satellite-Based Agricultural Intelligence

Repeated satellite observations support crop mapping, crop-type identification and growth monitoring. Multispectral, thermal and radar sensors capture complementary properties of crop condition, and temporal models such as convolutional temporal attention networks exploit the time dimension of image series for panoptic segmentation of agricultural parcels [1]. SICKLE, introduced by Sani et al. [2], is a multi-sensor dataset that combines Sentinel-1, Sentinel-2 and Landsat-8 imagery with crop labels and farming-calendar annotations for the Cauvery Delta of Tamil Nadu. It contains 2,370 season-wise samples from 388 plots and 21 crop types, and benchmarks crop-type mapping, sowing/transplanting/harvest-date prediction and yield prediction with a UNet3D architecture [25]. Its published late-fusion results include a paddy versus non-paddy IoU of 81.07% and a sowing-date MAE of 2.33 days, while yield MAPE remains between roughly 54% and 74% across sensors, showing that yield estimation is still difficult.

3.2 Crop Disease and Pest Recognition

Image-based plant-disease detection is a mature research area. The PlantVillage repository [3] contains more than 54,000 images of healthy and diseased leaves, and Mohanty et al. [4] reported 99.35% accuracy with deep convolutional networks on a held-out split covering 14 crop species and 26 diseases. Because PlantVillage images were captured under controlled conditions, such accuracy does not transfer directly to field photographs with varying light, angle and background. Field datasets address this gap: Paddy Doctor [5] provides 16,225 paddy-leaf images from fields near Tirunelveli in 13 categories, and several Indian field datasets for multi-crop, coconut, sugarcane, groundnut, black gram and rice diseases are available through Mendeley Data [6]–[11]. EfficientNetV2 [24] offers an efficient convolutional backbone suitable for transfer learning on such data. A recurring finding in these collections is the presence of duplicates and train–test leakage, which inflate reported accuracy unless removed.

3.3 Agricultural Knowledge Retrieval and Advisory Systems

Retrieval-augmented generation (RAG) conditions a language model on retrieved passages instead of relying only on parametric knowledge [12]. This suits agriculture, where answers must be traceable to documented guidance such as the TNAU Agritech Portal [13]. Multilingual dense retrievers such as BGE-M3 [14] enable Tamil and English retrieval with a single embedding model. Farmer.Chat [15] is a generative-AI assistant for smallholder farmers that its authors report has served more than 15,000 farmers and 300,000 queries in six languages across four countries. KisanQRS [16] maps farmers' queries to answers using Kisan Call Centre logs and reports query-mapping F1 of up to 96.58% and answer-retrieval NDCG of 96.20%. These systems focus on advisory question answering rather than on fusing evidence from several live data sources and models with explicit safety gates.

3.4 Weather, Soil and Market Information

Open services such as Open-Meteo [17] provide current weather and forecasts, and NASA POWER [26] provides historical meteorological data. SoilGrids 2.0 [18] produces global gridded soil-property estimates with quantified uncertainty, and the Soil Health Card programme [19] records field-level nutrient tests. AGMARKNET [20] publishes market arrivals and prices, while the Season and Crop Report of Tamil Nadu [27] and the ICRISAT District Level Database [28] provide crop-production statistics. An advisory system must label the difference between a regional soil estimate and a laboratory test, and between a historical price and a current one.

3.5 Speech Technology for Indian Languages

Whisper [21] showed that large weakly supervised models generalise across languages. For Indian languages, the Vistaar benchmark [22] reports that IndicWhisper achieves the lowest word error rate on 39 of 59 benchmarks, and the IndicVoices corpus [23] provides 7,348 hours of speech across 22 languages from 16,237 speakers in 145 districts. How well such models handle farm vocabulary and regional Tamil speech on low-end farmer devices remains an open question.

3.6 Summary of Literature

Table 3.1: Summary of related work

Ref.	Work	Contribution	Limitation relevant to this project
[2]	SICKLE (Sani et al., 2024)	Multi-sensor Cauvery Delta dataset; crop-type, calendar and yield benchmarks	Regional (Cauvery Delta); yield validated for paddy only; not farmer-facing
[3], [4]	PlantVillage; Mohanty et al.	Large disease dataset; 99.35% CNN accuracy	Controlled images; accuracy does not transfer to field photos
[5]	Paddy Doctor	16,225 field paddy images, 13 classes	Single crop; access and licence conditions apply
[12]	RAG (Lewis et al.)	Grounds generation in retrieved passages	Retrieval presence does not prove claim-level support
[15]	Farmer.Chat	Generative-AI advisory deployed at scale	Advisory QA focus; limited multi-source evidence fusion
[16]	KisanQRS	Query-to-answer mapping from call-centre logs	Retrieval of past answers; no live weather/market context
[22], [23]	Vistaar; IndicVoices	Indian-language ASR benchmarks and corpus	Not evaluated on farm vocabulary or low-end devices

3.7 Research Gap

Published agricultural AI work reports strong results in individual areas, but these components are usually built and tested as separate tools. What remains open is a single platform that combines Tamil and Tanglish interaction, source-linked knowledge retrieval, image-based disease screening, weather-aware advice, satellite crop intelligence, market and scheme information, and farm-specific context, while handling uncertainty explicitly and keeping

evidence visible. UZHAVAN AI treats this as an **integration and validation gap**: it does not claim that each individual model or data source is novel, but connects them through a common workflow that accounts for intent, farm context, evidence provenance, uncertainty and safety.

4. REQUIREMENT ANALYSIS

The requirements below are derived from the project's Software Requirements Specification (UZH-SRS-001, version 1.0, 25 October 2026). Each requirement has an identifier, a priority and an observable acceptance criterion. Targets describe desired behaviour and are not claims that the behaviour is already verified.

4.1 Operating Environment

- Android smartphone running the Flutter application (target test device: Oppo A9 2020, Android 11, API 30).
- FastAPI backend reachable over HTTPS in deployment; USB ADB-reverse loopback HTTP is used only for development.
- PostgreSQL, Redis and ChromaDB services, and an Ollama model service subject to model availability and resource capacity.
- External data providers subject to connectivity, licensing and rate limits.

4.2 Hardware and Software Requirements

Table 4.1: Hardware and software requirements

Category	Requirement
Client hardware	Android smartphone with microphone, speaker and camera / image picker
Server hardware	Development workstation capable of running a 7B-parameter LLM locally; GPU optional for disease inference
Mobile software	Flutter / Dart, Riverpod, go_router, Android TextToSpeech and on-device speech recogniser (Android 12+)
Backend software	Python, FastAPI, Uvicorn, PostgreSQL, Redis, ChromaDB
AI runtime	Ollama serving Qwen2.5 7B Instruct; PyTorch / torchvision for EfficientNetV2-S
External data	Open-Meteo, AGMARKNET-related market data, SoilGrids, TNAU corpus, official scheme sources

4.3 Functional Requirements

Table 4.2: Functional requirements

ID	Requirement	Priority
FR-01	Use the existing OTP/JWT flow to authenticate before accessing protected farmer data.	High
FR-02	Maintain farmer and farm context (language, district, crop, crop age) where supplied.	High
FR-03	Accept Tamil, Tanglish and English text questions and return a readable response.	High
FR-04	Capture speech on user activation and display a transcript from the speech service.	High
FR-05	In voice mode, auto-submit exactly one non-empty final transcript.	High

ID	Requirement	Priority
FR-06	Speak the final response in Tamil or English when a compatible voice is available.	High
FR-07	Show Listening, Thinking and Speaking states; stop audio on Stop or new microphone session.	High
FR-08	Classify intent and invoke only the required capabilities.	High
FR-09	Retrieve relevant TNAU and trusted documents before generating a grounded response.	High
FR-10	Expose available source title and retrieval time; never fabricate a citation.	High
FR-11	Allow labelled general LLM guidance for low-risk questions when retrieval finds nothing useful.	High
FR-12	Accept a leaf image and, when the approved model is enabled, return class and confidence.	High
FR-13	Return UNSURE when confidence is below the configured threshold.	High
FR-14	Separate model prediction from guidance; do not invent cause, symptoms or treatment.	High
FR-15	Retrieve weather and forecast data with location and freshness information.	High
FR-16	Retrieve market information with source, period and unit.	Medium
FR-17	Present regional soil context distinctly from measured farm-specific soil results.	Medium
FR-18	Withhold satellite predictions until licence, unit, scope and evaluation gates pass.	High
FR-19	Retrieve scheme information from official sources; do not assert unverified eligibility.	Medium
FR-20	Fuse capability outputs with provenance and apply grounding and safety checks.	High
FR-21	Preserve multi-turn context (crop, age, symptoms) within a conversation.	High
FR-22	Never return an exact pesticide/chemical dose without trusted, applicable evidence.	Critical
FR-23	Allow switching from voice to text, stopping speech and retrying without duplicates.	High
FR-24	Show actionable, non-technical feedback for network, provider, model and TTS failures.	Medium

4.4 Non-Functional Requirements

Table 4.3: Non-functional requirements

ID	Attribute	Requirement / target
NFR-01	Response time	95% of text-only responses begin rendering within 5 s in the agreed test environment.
NFR-02	Voice responsiveness	Auto-submission within 1 s of the final STT event for 95% of utterances.

ID	Attribute	Requirement / target
NFR-03	TTS start-up	TTS starts within 2 s of the final answer when a Tamil voice is installed.
NFR-04	Reliability	A 30-minute scripted demo completes without crash, duplicate submission or overlapping TTS.
NFR-05	Security	TLS for deployed traffic; no secrets or full tokens in logs.
NFR-06	Privacy	Collect only needed data; microphone and camera access user-initiated.
NFR-07	Safety	Chemical recommendations source-backed; low-confidence disease output never definitive.
NFR-08	Explainability	Distinguish observed, calculated, predicted and simulated information; show sources.
NFR-09	Usability	Core tasks completable without technical terminology.
NFR-10	Language support	Correct Tamil rendering; ta-IN TTS; unavailable language reported, not silently replaced.
NFR-11	Maintainability	Adapters, retrieval, orchestration, UI and speech remain separable and tested.
NFR-12	Interoperability	Provider failures or schema changes do not crash unrelated capabilities.
NFR-13	Auditability	Structured timestamps and correlation IDs without logging secrets.
NFR-14	Accessibility	Readable contrast, large tap targets and text alternatives for audio states.

4.5 External Interface Requirements

Table 4.4: Software interfaces

Interface	Purpose / requirement
Flutter → FastAPI	Authenticated request/response using existing token handling
FastAPI → PostgreSQL	Persistent farmer, farm, diary and conversation records
FastAPI → Redis	Caching and short-lived state where configured
Retrieval → ChromaDB	Vector search over the prepared TNAU knowledge collection
Assistant → Ollama	Natural-language generation with bounded, validated context
Weather adapter → provider	Current conditions and forecast with timestamp and location
Market adapter → source	Observed values with unit and date; never presented as live if a snapshot
Satellite adapter → catalogue	Remains gated until licence, units, area of interest and evaluation pass

The mobile user interface provides Home, Ask AI, Scan, Weather, Market, Soil, Schemes, Farm and Profile entry points; responses distinguish sources, confidence and uncertainty; and voice mode exposes distinct Listening, Thinking and Speaking states with a Stop control. Deployed client–server traffic uses HTTPS, and provider failures, timeouts and malformed responses are mapped to clear application-level errors.

4.6 Data Requirements

The main logical entities are Farmer/Account, Farm Profile, Farm Diary Entry, Conversation and Turn, Evidence Record, Assistant Response, Disease Screening, Weather/Market/Soil records and Satellite evidence. Input types, image dimensions and provider payloads are validated before processing; retrieval timestamps and source identifiers are attached to evidence; historical training records are never treated as current farm conditions; and source metadata is never manufactured.

4.7 Constraints and Business Rules

1. **Evidence-first generation:** retrieved or calculated evidence precedes the final LLM explanation when domain facts are required.
2. **No fabricated citations:** a source is shown only if a matching retrieved or provider record exists.
3. **Dose safety:** an exact chemical or pesticide dose requires trusted, context-applicable evidence.
4. **Confidence control:** low-confidence image predictions are never presented as definitive diagnoses.
5. **Satellite gate:** licence, units, geographic scope and independent evaluation must all pass before farmer-facing prediction.
6. **Provenance distinction:** observed, calculated, predicted and simulated values are not silently conflated.
7. **Authentication and model freeze:** assistant changes must not alter the OTP/JWT implementation or the M4 model registry without separate authorisation.

5. PROPOSED METHODOLOGY AND SYSTEM DESIGN

5.1 System Architecture

UZHAVAN AI follows a layered client–server architecture (Figure 5.1). The Flutter Android application communicates with a FastAPI backend over an authenticated REST API. The backend contains the authentication layer, API routes and the orchestration logic (intent router, planner and capability resolver). Three groups of services sit behind it: farm data (PostgreSQL and Redis), AI and retrieval (Ollama with Qwen2.5 7B, ChromaDB and the TNAU corpus), and domain services (weather, market, soil, schemes, the gated disease model and the gated satellite module). Their outputs pass through evidence fusion and grounding and safety validation before a farmer-facing response is returned.

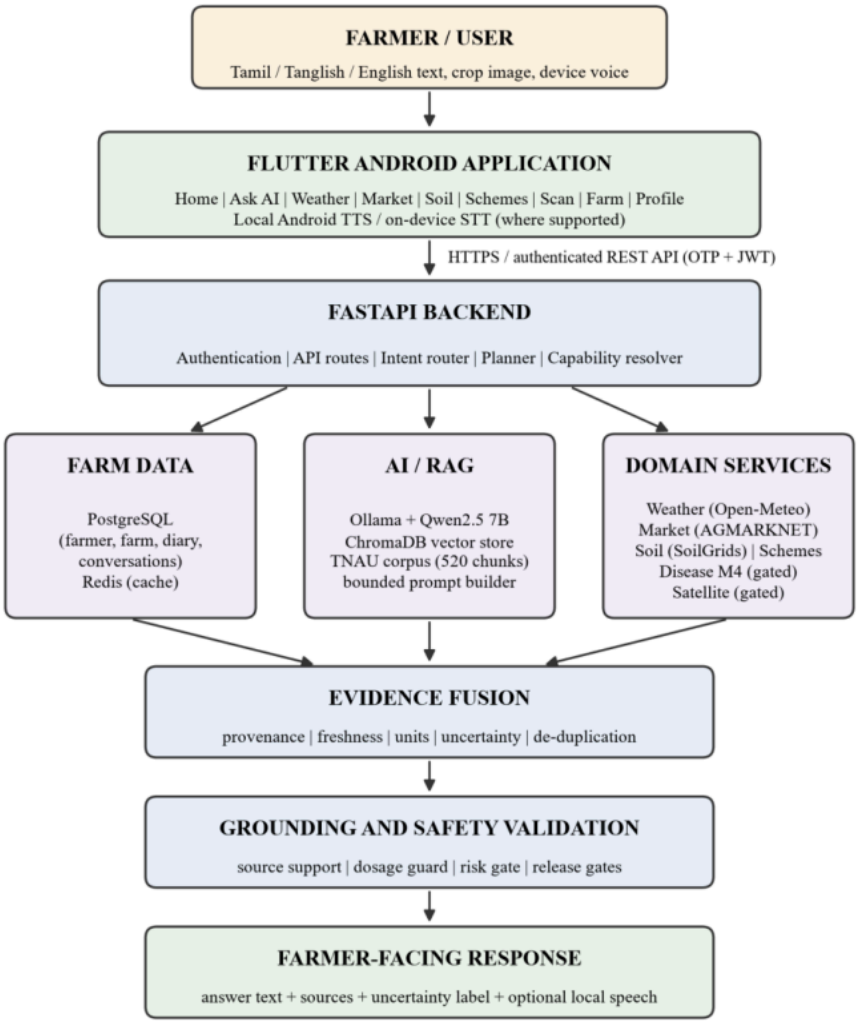


Figure 5.1: High-level architecture of UZHAVAN AI

5.2 Evidence-First Processing Pipeline

Every request passes through the same staged pipeline (Figure 5.2 and Table 5.1). Voice input is converted to text and follows the same path as typed input, so the voice channel adds no separate reasoning logic.

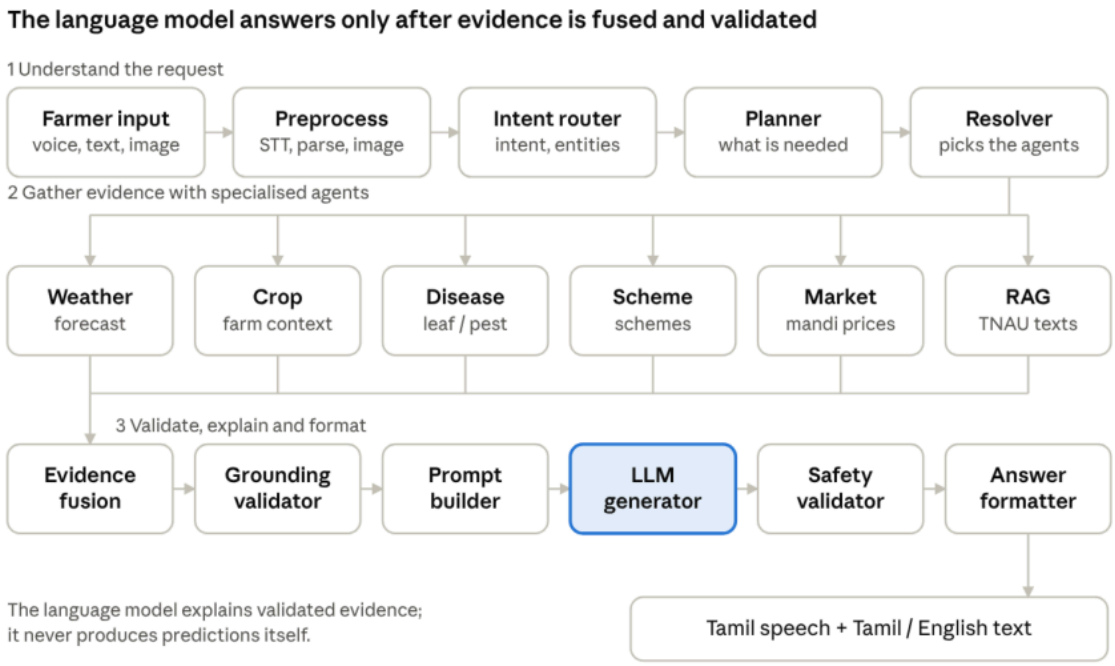


Figure 5.2: Evidence-first pipeline – the LLM is reached only after evidence is fused and validated

Table 5.1: Stages of the processing pipeline

Stage	Name	Responsibility
1	Farmer input	Accept a text question, supported voice transcript or crop image
2	Pre-processing	Normalise text, identify language, validate files and input size
3	Intent router and planner	Identify intent and entities such as crop, location and time
4	Capability resolver	Select only the domain services needed for the request
5	Domain executors	Retrieve TNAU content or obtain weather, market, soil, scheme or model evidence
6	Evidence fusion	Normalise, rank and de-duplicate evidence while retaining provenance and timestamps
7	Grounding and safety	Check support, relevance, freshness, uncertainty and risk constraints
8	LLM explanation	Local Qwen2.5 7B explains the validated evidence
9	Response formatting	Return readable text, actual source records and optional local speech

5.3 Design Principles

1. **Evidence before explanation:** facts are retrieved or computed before the LLM is asked to explain them.
2. **The LLM explains; it does not originate measurements:** weather, prices, soil values and model classes come from their source systems.

3. **Provenance is preserved:** sources and retrieval timestamps travel with the evidence.
4. **Uncertainty is explicit:** unsupported answers are labelled as general guidance, unsure or unavailable.
5. **Risk-aware fallback:** low-risk questions may receive labelled general guidance; exact dosage advice requires applicable evidence.
6. **Historical/current separation:** historical data trains and evaluates models; current provider data and farm context describe the present.
7. **Fail closed for gated capabilities:** disease and satellite release controls are never bypassed.
8. **Isolation of failures:** a provider or model timeout must not crash unrelated modules.

5.4 Formal Evidence Model and Response Policy

Each evidence item is represented as a tuple of its value v, source s, type κ (observed, calculated, predicted or retrieved), timestamp t and confidence c:

$$e_i = (v_i, s_i, \kappa_i, t_i, c_i) \tag{5.1}$$

An item is valid when it is confident enough, fresh enough for its type and comes from a trusted source:

$$valid(e_i) \Leftrightarrow c_i \geq \theta_c \ \wedge \ (t_now - t_i) \leq \Delta_\kappa \ \wedge \ s_i \in S_trusted \tag{5.2}$$

The language model G receives only the validated evidence set E* together with the query q and farm context f:

$$E^* = \{ e_i \in E : valid(e_i) \}, \quad answer = G(q, f, E^*) \tag{5.3}$$

The response then depends on the evidence and on the risk of the question:

- If E* is not empty, the answer is built from E* and carries its provenance and uncertainty.
- If E* is empty and the question is low risk, a bounded fallback answer is given and marked as general guidance.
- If E* is empty and the question requires an exact chemical or pesticide dose, the dose is withheld and the farmer is directed to the TNAU help centre or an agricultural officer.

5.5 Module Design

5.5.1 Authentication and farm context

An OTP/JWT workflow establishes the farmer session; protected endpoints reject requests without a valid Bearer token, and expired tokens are renewed through a refresh endpoint. The farmer profile stores preferred language, district, farm, crop, crop stage and land area, which are attached to a request only when relevant. A farm diary records activities such as land preparation, sowing, irrigation, fertiliser application and observed symptoms.

5.5.2 AI agricultural assistant (TNAU RAG)

The chat screen sends each turn to the backend, where the intent logic and planner select retrieval and other capabilities. TNAU documents are split into chunks, embedded and stored in a ChromaDB collection; at query time relevant chunks are retrieved with their source metadata. The local Qwen2.5 7B Instruct model, served by Ollama, prepares an explanation from this

bounded context, and the grounding and safety layer labels the response mode (for example grounded_verified or general_guidance) before the answer, status and sources are returned.

5.5.3 Weather, market, soil and scheme modules

The weather module presents current conditions and forecasts from Open-Meteo with their timestamps, separating current from forecast values. The market module shows commodity, market, price, unit, observation date and source, preserves the provider's Rs/quintal unit and labels snapshot data as "Snapshot – not live". The soil module distinguishes regional estimates (e.g. SoilGrids) from laboratory measurements of the farmer's field. The schemes module displays actual scheme names and source links and shows eligibility only when a reliable source returns it.

5.5.4 Disease screening (M4 research workstream)

The disease module uses an EfficientNetV2-S network fine-tuned by transfer learning on field-oriented leaf images, served by a separate inference service exposing /health and /predict. Classification is selective: for class probabilities $p_k(x)$ of an image x and a configured threshold θ_d, the module returns UNSURE rather than forcing a label when the model is not confident:

$$\hat{y}(x) = \text{argmax}_k\, p_k(x) \;\; \text{if} \;\; \max_k p_k(x) \geq \theta_d \;; \;\; \text{otherwise} \;\; \text{UNSURE} \tag{5.4}$$

In production-safe mode the farmer-facing route returns HTTP 503 ("The disease model is not yet approved for use") until the model passes external validation; a development-only research path returns outputs explicitly labelled as research demonstrations and not production-certified.

5.5.5 Satellite crop intelligence

The satellite module builds on the SICKLE dataset (Sentinel-1, Sentinel-2 and Landsat-8 over the Cauvery Delta) for crop type, crop-calendar and yield tasks. A farmer-facing satellite prediction is released only when all four gate conditions hold:

$$R_sat = G_licence \cdot G_units \cdot G_scope \cdot G_eval \;, \quad \text{released only if } R_sat = 1 \tag{5.5}$$

5.5.6 Voice interaction

Spoken responses use the native Android TextToSpeech engine with installed voice data; if a Tamil voice is not installed, the application states this and shows installation guidance instead of silently switching language or using a cloud voice. Speech input uses Android's on-device recogniser API where the platform supports it; typed input is always available.

5.6 Logical Data Model

Table 5.2: Logical data entities

Entity	Example attributes
Farmer	farmer_id, language preference, profile metadata
Farm	farm_id, farmer_id, district, crop, area, crop stage
FarmDiaryEntry	entry_id, farm_id, date, activity, note, image reference
Conversation / Message	conversation_id, role, text, timestamp, response status
KnowledgeChunk	chunk_id, title, text, source URI, page/section metadata
MarketObservation	commodity, market, price, unit, observation date, source
SoilRecord	location, property, value, unit, source, measurement type

Entity	Example attributes
SchemeRecord	scheme_id, title, eligibility text, source URI, last updated
DiseaseScreening	crop, class, confidence, model version, gate state
ValidationGate	capability, gate name, status, evidence reference

5.7 API Design

Table 5.3: Main backend API groups

Group	Endpoints
Authentication	/api/auth/send-otp, /verify-otp, /refresh, /logout
Farmer and farm	/api/farmer/profile, /api/farms, /api/farms/{farm_id}, /api/diary
Conversation and voice	/conversation, /voice/stt, /voice/tts
Disease screening	/api/disease/scan
Weather	/api/weather/current, /forecast, /advisories
Market	/api/market/commodities, /markets, /latest, /trend
Soil and schemes	/api/soil, /api/soil/tests, /api/schemes, /api/schemes/{id}/eligibility
Metadata and satellite	/api/meta/crops, /districts, /enums, /api/satellite/status
Health and operations	/health, /health/config, /metrics

6. IMPLEMENTATION

6.1 Technology Stack

Table 6.1: Technology stack

Layer	Technology	Role
Mobile app	Flutter / Dart (Riverpod, go_router)	Android UI, navigation, API integration, image input, voice controls
Backend	FastAPI / Python	Authenticated APIs, orchestration, domain routes, safety controls
Database	PostgreSQL	Farmer, farm, diary and conversation records
Cache	Redis	Caching and transient state
Vector store	ChromaDB	Semantic retrieval over the TNAU corpus (520 chunks)
LLM runtime	Ollama + Qwen2.5 7B Instruct	Local natural-language explanation; no cloud LLM
Disease vision	PyTorch / torchvision, EfficientNetV2-S	Experimental crop-leaf classification (research-only)
Speech	Android TextToSpeech; on-device recogniser	Local spoken output and speech input where supported
Weather / market	Open-Meteo; AGMARKNET-related sources	Forecast context; commodity observations in Rs/quintal
Satellite research	SICKLE (Sentinel-1/2, Landsat-8)	Crop and crop-calendar intelligence (gated)

6.2 Project Structure and Runtime Services

The code base is organised into separate workspaces: uzhavan_flutter (mobile application), uzhavan_platform (FastAPI backend, routes, services and tests), uzhavan_ml (M4 disease model and inference server) and uzhavan_data (curated datasets). During development the services run locally as listed in Table 6.2; a physical phone reaches the backend through ADB reverse port forwarding with a debug-time API URL override.

Table 6.2: Local runtime services

Service	Port	Purpose
FastAPI backend	8000	Mobile and local API requests
M4 inference server	8001	Research-only crop model inference (/health, /predict)
PostgreSQL	5432	Structured persistence
Redis	6379	Cache / transient state
Ollama	11434	Local LLM runtime

6.3 Backend Implementation

The FastAPI backend implements the authentication routes (OTP send/verify, token refresh, logout), farmer/farm/diary routes, the conversation route, the domain routes for weather, market, soil, schemes, metadata and satellite status, and health/metrics endpoints. The conversation route runs intent detection and planning, calls the TNAU retriever and the relevant adapters, applies evidence fusion, invokes Qwen2.5 7B through Ollama with bounded context, and passes the draft through the grounding and dosage guards. A /health/config endpoint reports whether the LLM provider, database, Redis, ChromaDB and disease-service URL are configured and reachable, so that an HTTP 200 from /health is not mistaken for full system health.

6.4 Retrieval and Safety Layer

TNAU agricultural content was chunked and loaded into a ChromaDB collection holding 520 chunks (English at the latest audit), with source title and page metadata retained. Retrieved chunks are passed to the prompt builder together with the farm context. The safety layer enforces the dosage guard: when a draft contains a chemical rate that is not present in applicable evidence, the rate is blocked and the farmer receives a safe fallback directing them to the TNAU help centre (0431-2371622) and an agricultural officer. Response labels such as grounded_verified and general_guidance tell the user whether the answer is supported by retrieved sources.

6.5 Mobile Application

The Flutter application uses Riverpod for state management and go_router for navigation in a five-tab shell, with screens for Home, Ask AI, Weather, Market, Soil, Schemes, Scan, Farm, Login, Profile and Settings. The assistant screen renders message bubbles, suggested questions, a Sources section (shown only when the backend returns real source metadata) and voice controls. A Material 3 UI/UX redesign with an agriculture-inspired palette (forest and leaf green, warm neutral background, harvest-amber accent) and a reusable component kit (app_theme.dart, ui_kit.dart) was started in November 2026; the shell, Home, Weather and Assistant screens have been rebuilt, and the remaining screens are in progress.

6.6 Disease Screening Service

The M4 inference server (m4_inference_server.py) loads an EfficientNetV2-S checkpoint and accepts a multipart leaf image with an optional crop. The canonical groundnut experiment (V2.2) has three classes – Healthy/Abiotic (1,449 images), Rust (2,235) and Tikka Leaf Spot (2,236) – for 5,920 training images. Before training, near-duplicate images are clustered and whole clusters are assigned to a single split, so that no training image is near-identical to a test image; for a similarity measure sim and threshold δ:

$$\forall a \in D_train,\ b \in D_test :\ sim(a, b) < \delta \tag{6.1}$$

The backend's production route remains fail-closed (HTTP 503) while DISEASE_MODEL_APPROVED is false, and the research branch tags every output with demo_mode: true and production_certified: false.

6.7 Data Foundation

Table 6.3: Data assets used in the project

Dataset / source	Recorded size / content	Use
Tamil Nadu multicrop collection	21,875 images, ~30 classes	Field-relevant disease screening
PlantVillage [3]	54,305 images	Reference benchmark / pre-training
IP102	75,222 images	Pest images (academic-use licence)
Indian field disease sets [6]–[11]	Coconut, sugarcane, groundnut, black gram, rice	Disease-screening training and testing
Tamil Nadu crop-yield data	45,504 rows, 38 districts, 105 crops, 1990–91 to 2024–25	Regional crop context
IMD / Open-Meteo weather	District rainfall and temperature records	Historical context; current forecasts
TNAU knowledge corpus [13]	520 chunks in ChromaDB	Source-linked retrieval
SICKLE [2]	2,370 samples, 388 plots, 21 crops	Satellite crop and calendar research

The crop-yield table separates paddy into the Kuruvai, Samba and Navarai seasons, maps administrative district splits, and marks whole-year aggregate rows to prevent double counting; after this correction, rice totals matched published Tamil Nadu figures (7.09 Mt in 2024–25). Data-quality audits found duplicates in parts of the coconut collection, about 19% duplicate images in one rice source group, and near-identical images across IP102 training and test splits, which motivates the leakage-safe splitting described above.

7. RESULTS AND DISCUSSION

7.1 Software Test Results

Table 7.1: Latest recorded software test results

Test suite	Result	Interpretation
Backend (pytest)	343 passed, 0 failed	Recorded after the backend safety and RAG repairs, using an isolated copy of the Chroma collection
Flutter (widget/unit)	180 passed	Recorded at an intermediate UI-redesign checkpoint; to be re-run after all screens are complete
Protected endpoints	HTTP 401 without token	Expected behaviour; expired token correctly refreshed (/api/auth/refresh 200)
Disease route (production mode)	HTTP 503	Intended fail-closed behaviour while the model is unapproved

These results show that the integrated software behaves as specified at the API level. They are not measures of agricultural accuracy, Tamil language quality or advisory usefulness.

7.2 Assistant Evaluation

On the physical test device (Oppo A9 2020, Android 11), the English assistant answered four questions in approximately 19–27 seconds. A later exact-question backend dry-run produced the results in Table 7.2.

Table 7.2: Exact-question assistant dry-run

Question	Latency	Mode	Observation
What is NPK fertilizer?	16.4 s	grounded_verified	Correct N, P, K definition; retrieved pages (Paprika, Anthurium) were not strong support
How can farmers improve soil health?	22.4 s	grounded_verified	Mulching, erosion control and organic matter advised; one duplicated source entry
Why are rice leaves turning yellow?	25.6 s	general_guidance	No source list; without farm context the draft contained an unsupported paddy rate

7.3 Safety Guard Result

When the rice-yellowing question was repeated on the phone with a banana farm as the active context, the backend safety log recorded a drafted rate of 120 kg N, 50 kg P and 50 kg K per hectare that was not present in applicable evidence. The dosage guard blocked it and the farmer received the safe fallback: “Sorry, the information needed to answer this is not available right now. Please contact the TNAU help centre: 0431-2371622”, together with a warning that no verified source was found. This demonstrates the evidence-first policy working as designed; it is not a crop diagnosis.

7.4 Physical Device Testing

- Login, Home, Weather, Market, Soil and Government Schemes screens loaded and displayed source/data information in a walkthrough on the Oppo A9 2020.
- The Market screen showed prices in Rs/quintal with a "Snapshot – not live" label.
- English text-to-speech worked with an embedded on-device voice; the Stop action was processed by the software, but audible silence has not yet been acoustically confirmed.
- The Scan screen loaded and the production-safe API returned the intended 503 "under validation" state.
- On-device speech recognition is unavailable on this phone because the current implementation relies on an Android 12+ API; typed input remains the supported path, and no cloud fallback is used.

7.5 Published Satellite Baseline

The only quantitative accuracy figures available for the satellite workstream are the published SICKLE UNet3D benchmark results [2], reproduced in Table 7.3 (mean ± standard deviation over five seeds). They are task-level baselines from the dataset paper and must not be read as end-to-end UZHAVAN AI results.

Table 7.3: Published SICKLE UNet3D benchmark results [2]

Task (metric)	Landsat-8	Sentinel-2	Sentinel-1	Late fusion
Paddy vs non-paddy (IoU, %)	56.04 ± 5.84	78.12 ± 3.48	81.77 ± 6.60	81.07 ± 5.77
Sowing date (MAE, days)	2.66 ± 0.96	2.30 ± 0.61	3.61 ± 0.90	2.33 ± 0.64
Transplanting date (MAE, days)	6.20 ± 1.03	6.36 ± 2.16	7.23 ± 0.78	6.16 ± 1.77
Harvest date (MAE, days)	9.86 ± 0.74	8.83 ± 1.52	10.08 ± 0.56	10.75 ± 3.39
Yield, actual season (MAPE, %)	59.38 ± 14.75	73.59 ± 9.81	65.66 ± 16.24	64.56 ± 13.77

7.6 Disease Model Evaluation

The groundnut M4 V2.2 model recorded a three-seed macro-F1 of 0.8342 ± 0.1163, below the agreed release gate of 0.85, and an open-set F1 of 0.6898. Later V2.3 runs regressed relative to this baseline. The model therefore remains on HOLD (no-go) for farmer-facing use, and its outputs are treated only as research screening results.

7.7 Planned Evaluation

Table 7.4: Planned evaluation of UZHAVAN AI

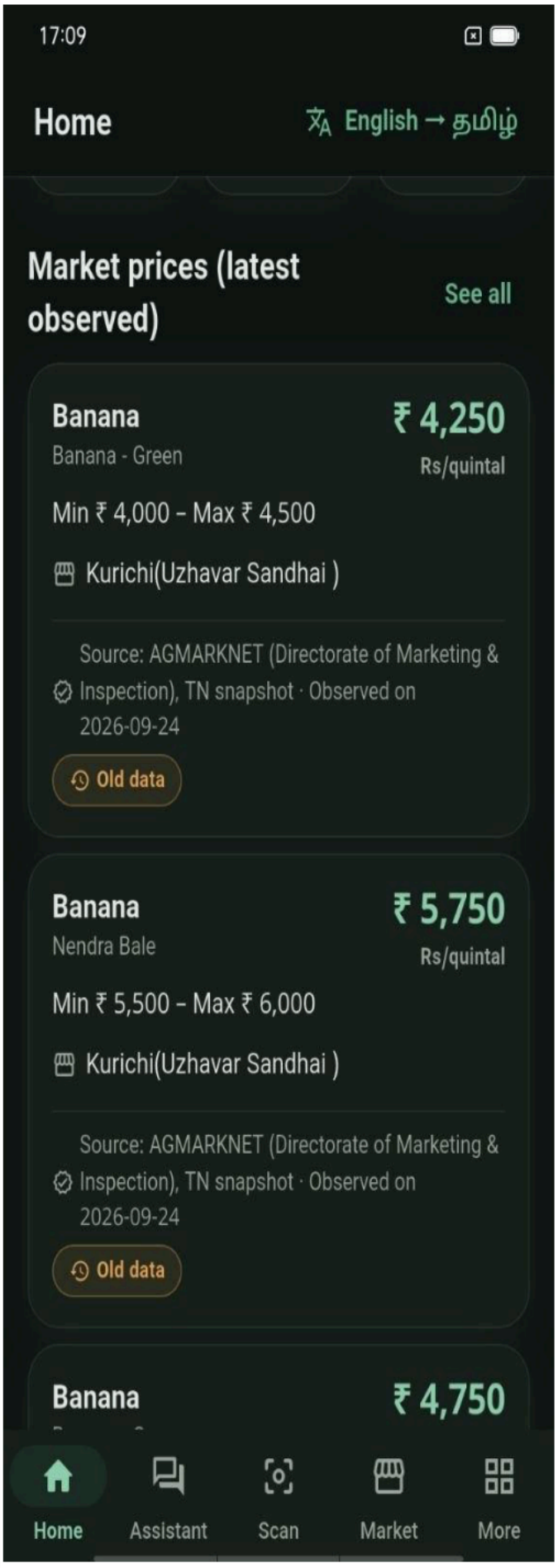
Component	Metric	Status
Tamil speech recognition	WER / CER (supported devices)	To evaluate
TNAU RAG	Recall@K, nDCG, grounded-answer rate	To evaluate
Disease classification	Macro-F1, per-class recall, ECE, coverage	Gate not yet passed

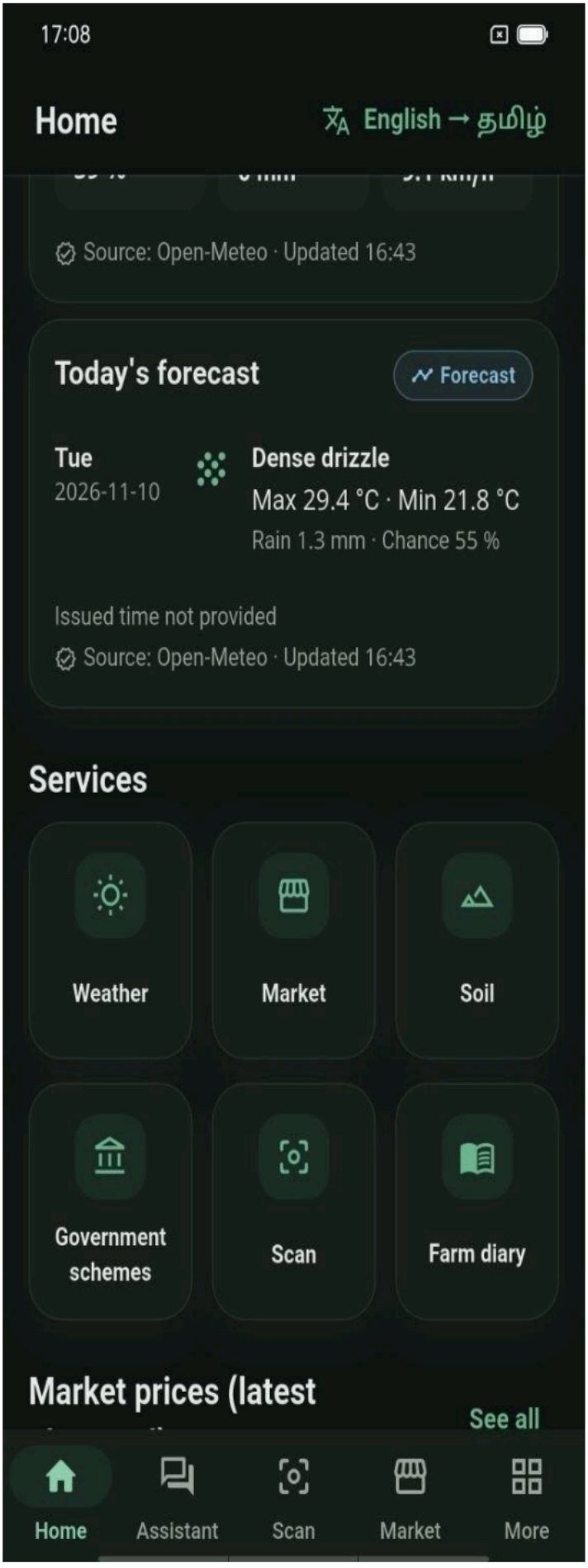
Component	Metric	Status
Satellite crop mapping	IoU (paddy vs non-paddy)	Published baseline 81.07%
Weather and market	Freshness, unit consistency, accuracy	To evaluate
End-to-end assistant	Task success, unsupported-answer rate	To evaluate
Farmer usability	Task completion, comprehension, trust	Planned field study

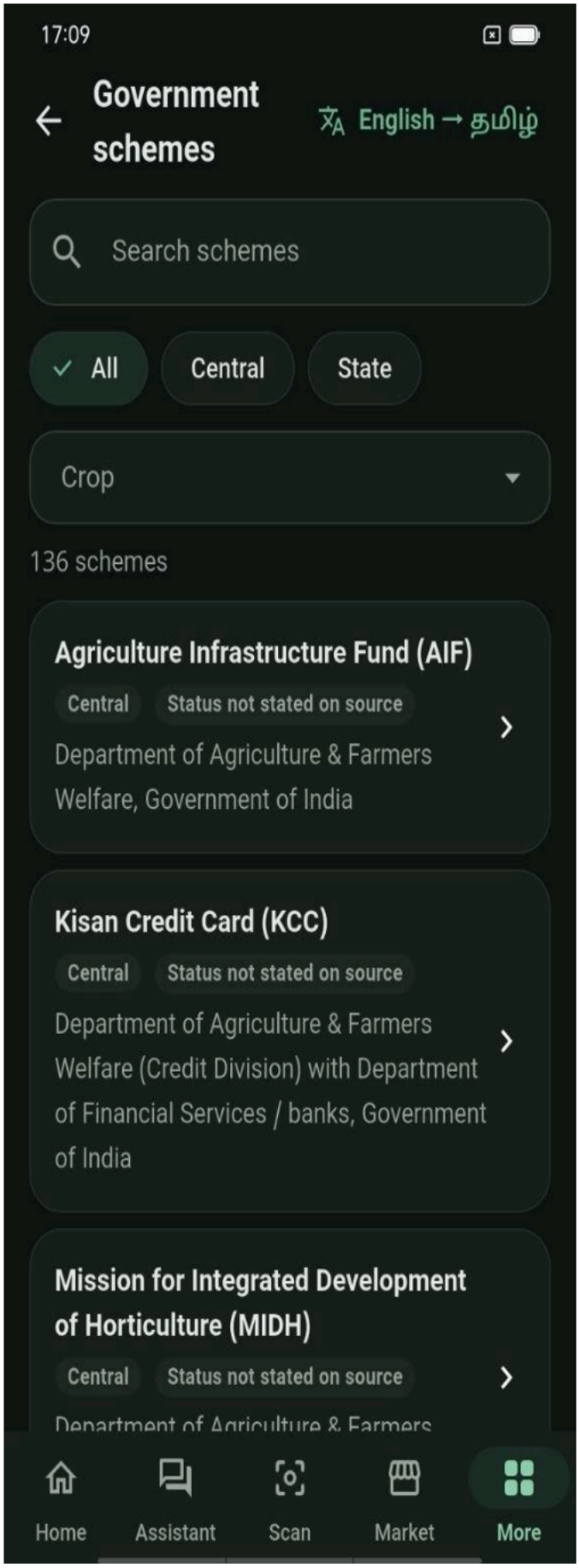
7.8 Discussion

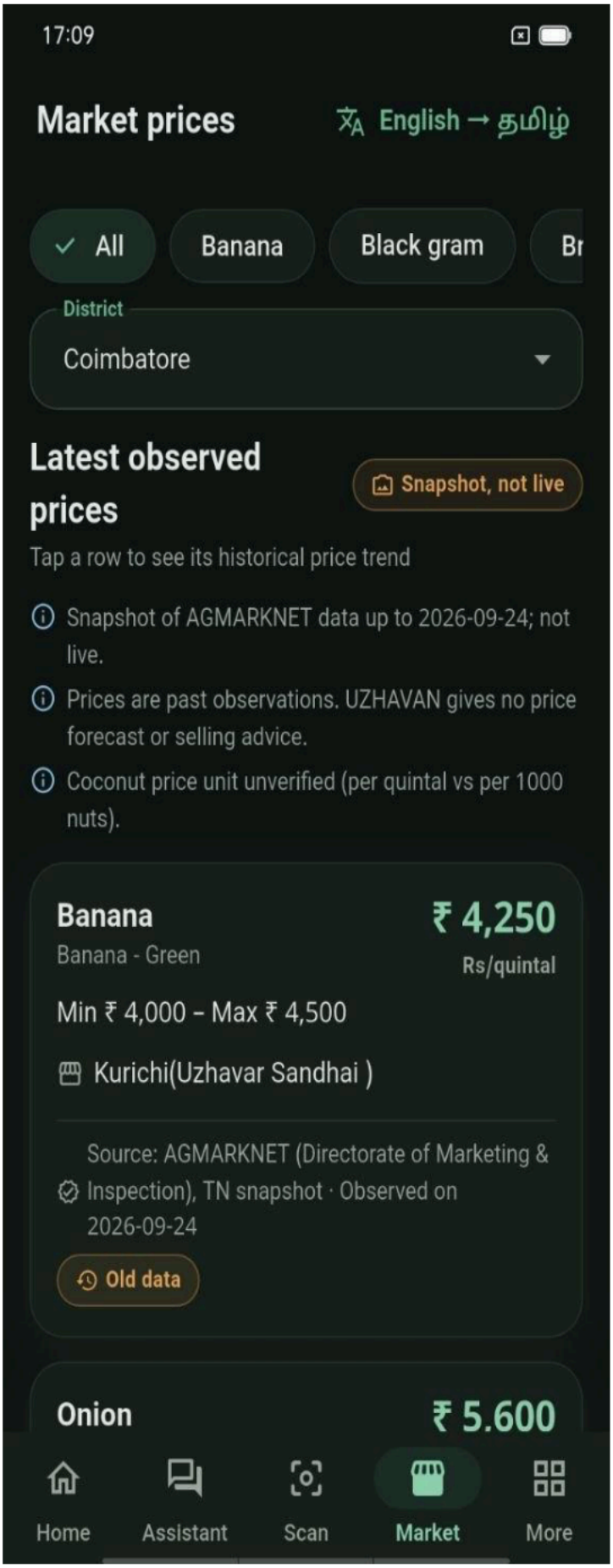
The results show that UZHAVAN AI provides a working technical base for combining several kinds of agricultural information in one mobile workflow, and that its safety design behaves as intended: unsupported fertiliser rates are blocked, the unapproved disease model stays behind a fail-closed gate, market snapshots are labelled as such, and source lists appear only when real metadata is returned.

Several points limit how far the results can be read. First, retrieval relevance must be evaluated at the claim level – in the NPK example the answer was correct but the retrieved documents did not clearly support it. Second, Tamil free-form chat is not yet of demonstration quality: earlier tests showed weather answers contradicting the Weather screen, prices expressed per kilogram instead of per quintal, and scheme queries routed to a generic path; the demonstrated interaction is therefore English-led. Third, the latest backend health check reported the PostgreSQL database as unreachable while Redis and ChromaDB were available, so farm-data availability on the phone must be re-verified. Fourth, satellite results are regional (Cauvery Delta) and yield is validated for paddy only. Finally, response latency of roughly 16–27 seconds on a local 7B model exceeds the 5-second NFR-01 target and needs optimisation.


17:09
Home
English → தமிழ்
Market prices (latest observed)
See all
Banana
Banana - Green
₹ 4,250
Rs/quintal
Min ₹ 4,000 – Max ₹ 4,500
Kurichi(Uzhavar Sandhai)
Source: AGMARKNET (Directorate of Marketing & Inspection), TN snapshot · Observed on 2026-09-24
Old data
Banana
Nendra Bale
₹ 5,750
Rs/quintal
Min ₹ 5,500 – Max ₹ 6,000
Kurichi(Uzhavar Sandhai)
Source: AGMARKNET (Directorate of Marketing & Inspection), TN snapshot · Observed on 2026-09-24
Old data
Banana
₹ 4,750
Home
Assistant
Scan
Market
More


17:08
Home
English → தமிழ்
Source: Open-Meteo · Updated 16:43
Today's forecast
Forecast
Tue
2026-11-10
Dense drizzle
Max 29.4 °C · Min 21.8 °C
Rain 1.3 mm · Chance 55 %
Issued time not provided
Source: Open-Meteo · Updated 16:43
Services
Weather
Market
Soil
Government schemes
Scan
Farm diary
Market prices (latest
See all
Home
Assistant
Scan
Market
More


17:09
Government schemes
English → தமிழ்
Search schemes
All
Central
State
Crop
136 schemes
Agriculture Infrastructure Fund (AIF)
Central
Status not stated on source
Department of Agriculture & Farmers Welfare, Government of India
Kisan Credit Card (KCC)
Central
Status not stated on source
Department of Agriculture & Farmers Welfare (Credit Division) with Department of Financial Services / banks, Government of India
Mission for Integrated Development of Horticulture (MIDH)
Central
Status not stated on source
Home
Assistant
Scan
Market
More


17:09
Market prices
English → தமிழ்
All
Banana
Black gram
District
Coimbatore
Latest observed prices
Snapshot, not live
Tap a row to see its historical price trend
Snapshot of AGMARKNET data up to 2026-09-24; not live.
Prices are past observations. UZHAVAN gives no price forecast or selling advice.
Coconut price unit unverified (per quintal vs per 1000 nuts).
Banana
Banana - Green
₹ 4,250
Rs/quintal
Min ₹ 4,000 – Max ₹ 4,500
Kurichi(Uzhavar Sandhai)
Source: AGMARKNET (Directorate of Marketing & Inspection), TN snapshot · Observed on 2026-09-24
Old data
Onion
₹ 5.600
Home
Assistant
Scan
Market
More

17:09

Crop disease scan

English → தமிழ்



Your photo

Retake | Remove

Send for checking

M4 v1 RESEARCH DEMO

NOT PRODUCTION CERTIFIED

Sigatoka Leaf Spot

Crop	BANANA
Confidence	89.9 %
Model	EfficientNetV2-S
Experiment	UZ-M4-v1.0-a898475e-s1

Home | Assistant | Scan | Market | More

17:08

Home

English → தமிழ்

Hello, Ajai

Tuesday, November 10

YOUR FARM

karunya

Crop	Crop stage	Land size (acres)
Banana	Land preparation	3.0 acres

Ask the farming assistant

Crops, soil, weather and schemes. Sources are shown when they are found.

Weather

See all

Current weather

Current

Location: karunya

28 °C

Overcast

Home | Assistant | Scan | Market | More

8. CONCLUSION

This report presented UZHAVAN AI, a Tamil-first agricultural intelligence and decision-support platform for farmers in Tamil Nadu. The platform integrates a Flutter Android application, a FastAPI backend, a TNAU-oriented retrieval layer over ChromaDB, local language generation with Qwen2.5 7B through Ollama, and modules for weather, market, soil and government-scheme information, together with research-stage disease-screening and satellite workstreams placed behind explicit release gates.

The main contribution is the evidence-first, capability-based architecture, in which the language model explains validated evidence instead of producing facts, provenance and uncertainty are kept visible, and high-risk outputs such as exact chemical doses and unvalidated disease classes are withheld. Software validation (343 backend tests and 180 Flutter tests passing) and physical-device tests confirm that the core workflow and its safety controls operate as designed. The project is, however, an actively developed prototype rather than a validated commercial diagnostic system.

8.1 Future Work

- Restore and verify database connectivity and complete the physical-device walkthrough of all redesigned screens.
- Improve Tamil and Tanglish answer quality and intent routing, add Tamil content to the retrieval corpus, and evaluate them with native speakers.
- Support speech input on Android 11 and lower devices using a local recogniser, without cloud fallback.
- Evaluate retrieval with an expert-reviewed query set (Recall@K, nDCG, claim-level support) and remove duplicate source entries.
- Re-train and externally validate the disease model on leakage-free, source-disjoint field data, with calibration and an UNSURE threshold.
- Reconcile satellite metrics with the original SICKLE checkpoints and validate beyond the Cauvery Delta before any farmer-facing release.
- Reduce response latency and conduct a farmer-centred usability and field study with informed consent.

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10. ANNEXURE - I: OUTCOME-SPECIFIC SUPPORTING EVIDENCE

Selected project outcome: Journal Publication.

A. Manuscript Details

Table A.1: Manuscript details

Item	Details
Manuscript title	UZHAVAN AI: A Tamil-First Multimodal Agricultural Intelligence and Decision-Support Platform for Tamil Nadu Farmers
Authors	Ajai Kumar N, Harish Gokul G, Dr. Samuel Nirmalsingh S (Karunya Institute of Technology and Sciences)
Target journal	Smart Agricultural Technology (Elsevier)
Article type	Full Length Article
Manuscript file	UZHAVAN_AI_IEEE_paper.docx
Current status	Submission in progress – manuscript uploaded and competing-interests declaration confirmed; title, abstract, keyword and author-detail steps pending; no journal receipt yet

B. Abstract of the Manuscript

Farming decisions in Tamil Nadu depend on evidence scattered across crop knowledge, weather, soil, disease imagery, satellite observations, market prices and government schemes, and most digital tools address only one of these. The manuscript presents UZHAVAN AI, a Tamil-first multimodal decision-support platform in which a staged pipeline performs intent classification, capability planning, execution of specialised agricultural agents, evidence fusion, grounding validation, language-model explanation and safety checking, so that the language model explains evidence instead of producing it. The satellite workstream builds on the SICKLE dataset, whose published benchmark is reported as an external baseline, and the implementation integrates a TNAU retrieval layer (520 chunks), local language generation and a gated disease-screening service. The paper states the limits of the present evidence and defines the planned evaluation of retrieval grounding, Tamil speech, disease classification and farmer-centred use.

C. Proposed Contributions

1. An evidence-first, capability-based orchestration architecture for agricultural decision support.
2. Integration of a TNAU-oriented retrieval layer with weather, market, soil and scheme services.
3. A farm-context-aware workflow that distinguishes retrieved, observed, calculated and predicted information.
4. Safety controls for unsupported chemical dosage and unapproved disease-model output.
5. A transparent account of what is measured today and a formally defined evaluation plan.

D. Pre-Submission Checklist

- Format the manuscript to the Smart Agricultural Technology guide for authors (not only the IEEE template).
- Update the software test results to the latest verified suite (343 backend and 180 Flutter tests) in place of the earlier 59/59 figure.
- Describe Tamil speech as a design goal with the documented Android 11 limitation, and the disease model as research-only (HOLD).
- Keep published SICKLE baselines clearly separate from UZHAVAN measurements.
- Verify citations, dataset licences and figure permissions, and add the generative-AI use declaration required by the journal.
- Confirm author list, order, affiliations, contributor roles and consent of every author.

Proof of submission (submission acknowledgement e-mail and manuscript number) will be attached to this annexure once the submission is completed.